miRNA-Disease Association Prediction based on Heterogeneous Graph Transformer with Multi-view similarity and Random Auto-encoder.

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***Abstract*—MicroRNAs (miRNAs) are a class of short non-coding single-stranded RNA molecules that play a key role in gene expression regulation. Understanding the association between miRNAs and diseases is crucial for disease diagnosis and treatment. Although the wet experimental methods can be used to determine the associations, they are both laborious and expensive. In this paper, we propose a novel computational method called TWMHGT for predicting the associations between miRNAs and diseases based a two-way Multi-layer Heterogeneous Graph Transformer (MHGT) framework. For the first way, multi-view similarity of miRNAs and diseases is used as the input encodings for MHGT, and for the second way, random auto-encoders is used to generate the input encodings. In each MHGT way, the encodings of each layer are concatenated to obtain comprehensive embeddings of miRNAs and diseases, thereby integrating multiple high-level information. Then, the attention mechanisms are used to fuse the embeddings generated from the two-way MHGT. During the decoding stage, the fused embeddings are decoded to obtain the predicted association matrix by using matrix multiplication. Our model was benchmarked on two datasets and the 5-fold and 10-fold cross-validation results show that TWMHGT outperforms the state-of-the-art methods in terms of AUC, AUPR, accuracy, sensitivity, and specificity. Furthermore, we conducted case studies on three different diseases to validate the predictive performance of TWMHGT. The results show excellent performance in all cases, indicating the potential of TWMHGT in discovering novel miRNA-disease associations.**

Keywords—miRNA-disease association, Graph Embedding, Multi-layer Heterogeneous Graph Transformer, Attention Mechanism

I. INTRODUCTION

MicroRNAs (miRNAs) are a class of short non-coding single-stranded RNAs which contain about 19-25 nucleotides [1]. miRNAs regulate the expression levels of proteins and genes by binding to the 3' untranslated region of target mRNAs [2-4]. Additionally, there is increasing evidence that miRNAs play critical roles in various biological processes such as cell proliferation, differentiation, apoptosis, aging, etc., and are closely related to the occurrence of many complex human diseases.Therefore, identifying the associations between diseases and miRNAs can help researchers to understand the mechanisms of various diseases, thereby benefiting for the detection, diagnosis, and treatment of complex diseases [5, 6].

Several wet lab methods have been used to detect the associations between miRNAs and diseases such as PT-PCR [7], Northern blotting [8]and microarray profiling [9]. However, these methods are laborious and expensive. With the development of computer technology, a bunch of computational methods have been developed to identify the associations between miRNAs and diseases, which can be categorized into two classes: similarity-based measurement [10-12] and machine learning-based methods [13-16].

Similarity-based methods are only based on the similarities between miRNAs or diseases which lead to the unsatisfied performance, so that machine learning-based models have been developed for predicting miRNA-disease associations. For example, You et al. proposed PBMDA model based on a heterogeneous graph that is composed of three interrelated subgraphs. The model used a depth-first search algorithm to obtain scores for miRNA-disease associations [13]. Xu et al. [14] built a heterogeneous miRNA-target dysregulation network (MTDN) to represent miRNAs and disease, and then used support vector machines to score the miRNA-disease associations. In addition, Chen et al. [15] proposed a method, namely LRSSLMDA, which uses Laplacian regularization and sparse subspace learning to learn associations between miRNAs and diseases. In another work, they used auto-encoders to obtain the embeddings of miRNAs and diseases and then using a random forest classifier to predict the associations [16].

In recent years, graph neural networks (GNN) [17] such as graph convolutional networks (GCN) [18] and graph attention networks (GAT) [19] have received widespread attention for their significant ability to handle graph-structured data. For example, Li et al. [20] used a graph convolutional network (NIMCGCN) to learn the latent feature representations of miRNA and disease, which are then embedded into NIMC for association prediction. Tang et al. [21] proposed MMGCN, which applied a multi-view multi-channel attention mechanism with graph convolutional network to enhance the features of miRNA and disease. Later, researchers noticed the rich semantic information in the miRNA-disease heterogeneous graph and then proposed relevant methods based on it. For example, the HGANMDA model constructs a miRNA-disease-lncRNA heterogeneous graph network to predict miRNA-disease associations [22]. Ning et al. [23] proposed a model called AMHMDA, which uses attention-aware multi-view similarity networks and hypernodes for calculating miRNA-disease association scores. Although the above methods have improved the performance for predicting miRNA-disease associations, they do not sufficiently integrate local information into the constructed network.

In this paper, we propose a computational method called TWMHGT, which is based on two way encoding modes and Multi-layer Heterogeneous Graph Transforms (MHGT), for miRNA-disease association prediction. The effectiveness of TWMHGT was evaluated on two different datasets using 5-fold and 10-fold cross-validation. Experimental results

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demonstrate that our method outperforms existing state-of-the-art approaches.

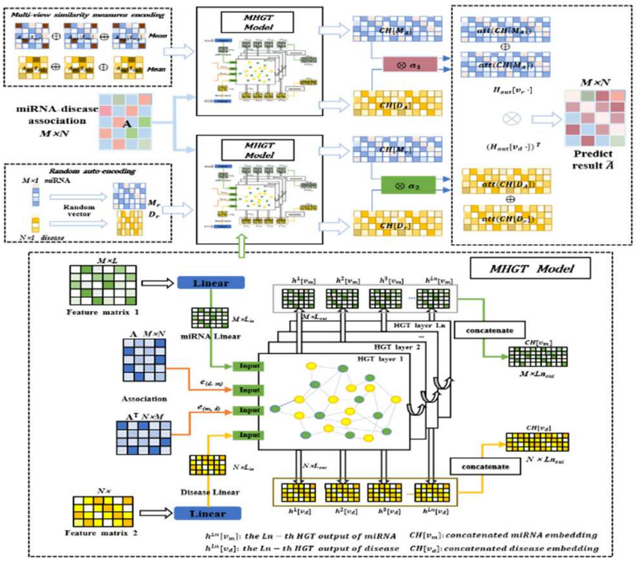

Fig. 1. The flowchart of TWMHGT.

II. MATERIALS AND METHODS

A. Data sets

Different data sets have been used in previous works, and these datasets are mainly collected from version 2.x or 3.x of Human MicroRNA Disease Database. To extensively validate our model, we benchmarked our method on two datasets. The first one is used in AMHMDA[23] , which includes 12,446 associations between 853 miRNAs and 591 diseases. We refer to this dataset as Data1. The second dataset is obtained from the work of MDA-CF[24], which includes 14,550 associations between 917 miRNAs and 792 diseases. We refer to this dataset as Data2. In the datasets, known miRNA-disease associations were marked as positive samples (1), while the remaining samples were denoted as negative samples (0), indicating unknown association between miRNAs and diseases. To ensure fairness in the prediction results, we randomly selected an equal number of negative samples as the positive samples to construct a balanced dataset.

B. TWMHGT model

The workflow of the proposed TWMHGT model is shown in Figure 1. The first encoding mode is based on the similarity between miRNA and disease from multiple views, while the second encoding mode is based on random auto-encoders. Then, the two sets of encodings are fed into two different MHGT networks to extract high-level embeddings separately. In MHGT, the output encodings from each layer of MHGT are concatenated to form the final encoding. The attention mechanism is then used to fuse the embeddings obtained from the two-way MHGT. Then the matrix multiplication was used to decode the fused embeddings so as to obtain the predicted miRNA-disease sensitivity association matrix. We used several evaluation metrics, such as area under receiver operating characteristic curve (AUC), area under the precision-recall curve (AUPR), accuracy, precision, recall, F1-score, and specificity, to evaluate the model performance. The details of our model can be found in the supplemental materials (https://github.com/yinboliu-git/TWMHGT).

III. RESULTS AND DISCUSSIONS

A. Experimental setup

To validate the effectiveness of the model, we conducted experiments on two datasets, Data1 and Data2, respectively. To maintain dataset balance, an equal number of negative samples were randomly selected as samples labeled 0. For optimal predictive performance, we employed 5-CV combined with grid search to adjust the model parameters and structure. The data was divided into training and test sets, with the test set accounting for 20% of the dataset. In our experiments, we set the epoch to 400, the learning rate to 0.01, weight decay to 0.0002, and dropout rate to 0.5 to address potential overfitting issues. These hyperparameters were determined through multiple trials and comparisons in order to achieve the best training performance. These findings provide important insights into the design and development of HGT models for predicting miRNA-disease associations.

TABLE 1. PERFORMANCE COMPARISON BASED ON 5-CV EXPERIMENTS OF DATA1.

Method	AUC	AUPR	ACC	F1-score	Recall	Precision
TWMHGT	**0.9677**	**0.9604**	**0.9149**	**0.9171**	**0.9417**	**0.8938**
HGANMDA	0.9265	0.9253	0.8489	0.8481	0.8433	0.8529
AGAEMD	0.9270	0.9286	0.8502	0.8507	0.8544	0.8481
MINIMDA	0.9304	0.9350	0.8481	0.8482	0.8529	0.8505
MAGCN	0.9245	0.9268	0.8483	0.8473	0.8425	0.8533
AMHMDA	0.9422	0.9411	0.8669	0.8653	0.8549	0.8763

TABLE 2. PERFORMANCE COMPARISON BASED ON 5-CV EXPERIMENTS OF DATA2

Method	AUC	AUPR	ACC	F1-score	Recall	Precision
TWMHGT	**0.9710**	**0.9647**	**0.9201**	**0.9221**	**0.9457**	**0.8998**
MDA-CF	0.9464	0.9461	0.8767	0.8759	0.8696	0.8859
MNGACDA	0.9516	0.9485	0.8672	0.8847	0.8823	0.8616
GATECDA	0.9413	0.9420	0.8581	0.8727	0.8705	0.8528
MINIMDA	0.9420	0.9405	0.8677	0.8694	0.8808	0.8585

B. Parameter sensitivity analysis

We conducted parameter sensitivity analysis to analyze how the model parameters affect the performance. The TWMHGT model includes the following parameters: (1) Hidden Channels: the dimension of hidden layer output encoding in HGT (The search range was 64, 128, 256, 512.), (2) Num Heads: the number of heads in the multi-head attention mechanism in HGT (The search range was 2, 4, 8,16.), (3) Num Layers: the number of layers in the multi-layer HGT (The search range was 2, 4, 6,8.), (4) RAES Dim: the dimension of the random auto-encoders (The search range was 64, 128, 256, 512.).

As shown in Figures in supplemental materials, the 5-CV results on data1 show that the optimal performance was achieved when Hidden Channels, Num Heads, Num Layers are set to 512, 8, 2 and 256, respectively. For Data2, when Hidden Channels, Num Heads, Num Layers and RAEs Dim are set to 512, 8, 4, and 256, respectively, the optimal performance is achieved.

A. Comparison with other methods

In this study, we compared our model with five state-of-the-art methods on Data1, including HGANMDA, AGAEMD, MINIMDA, MAGCN and AMHMDA. Table 1 shows the 5-

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CV performance of different models on Data1. In addition, we compared several methods including MDA-CF, MNGACDA, GATECDA, MINIMDA, and the results are shown in Table 2. The two tables clearly show that our model consistently outperforms other models on multiple datasets.

B. *Ablation tests*

The TWMHGT model consists of four key modules: multi-view similarity measurement, random encoding, MHGT, and element-wise multiplication decoding module. To evaluate the impact of each component, we conducted 5-CV experiments on benchmark datasets by removing each module. Specifically, we tested and compared the following six models: **TWMHGT-MD** model: This model does not use the element-wise multiplication operation but employs linear layers for decoding. **TWMHGT-RAEs** model: The TWMHGT-RAEs model does not use random encoding. **TWMHGT-HGT** model: This model does not use HGT as the core algorithm but adopts HAN for core encoding. **TWMHGT-Linear** model: This model removes the linear layer that comes before MHGT. **TWMHGT-CM** model: This model retains the random encoding module and MHGT framework but only uses the last layer of MHGT as the output embedding.

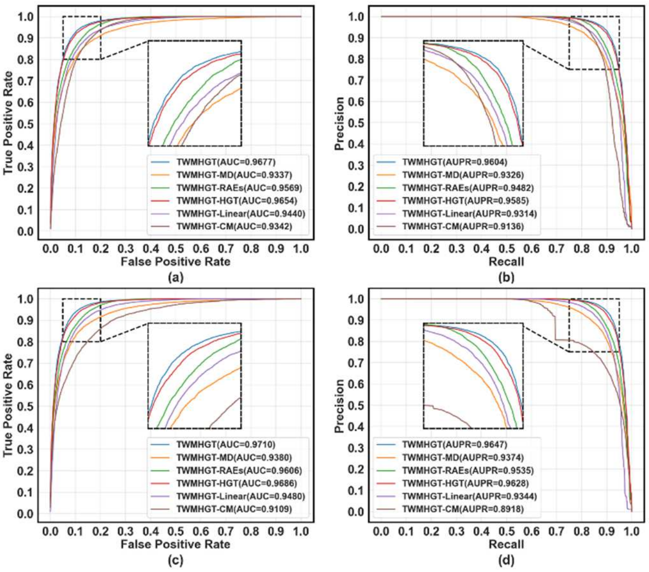

Fig. 2. The ROC and PR curves for different ablated models based on 5-CV on Data1 (upper panels) and Data2 (lower panels)

Figure 2 shows that the all components contribute to the final model. Specifically, the embeddings from different HGT layers are the most important, which demonstrate the complementarity between the embeddings from different HGT layers.

C. *Case studies*

To further demonstrate the performance of TWMHGT in identifying novel potential miRNA-disease association prediction in real cases, we conduct case studies on three diseases, which are colon neoplasms, gastric neoplasms and breast neoplasms as case studies. Specifically, in each case study, to prove the ability of TWMHGT in predicting new diseases without any known associated miRNAs, we utilized all known associations obtained from HMDDv3.2[25] to build the model, but eliminate all relevant known disease information from the dataset. Subsequently, the score matrix is obtained through this model. Afterwards, we sorted the predicted results and selected the top 20 candidate miRNAs predicted by TWMHGT. We used the authoritative dataset dbDEMC 3.0 [26] as a validation set to determine whether the predicted miRNAs were confirmed to be associated with specific diseases.

TABLE 3. THE TOP 20 MIRNAS PREDICTED BY TWMHGT FOR COLON NEOPLASM.

Colon neoplasm					
Rank	**miRNA**	**Evidence**	**Rank**	**miRNA**	**Evidence**
1	hsa-mir-4258	HMDD v3.2	11	hsa-mir-10a	HMDD v3.2
2	hsa-mir-3186	HMDD v3.2	12	hsa-mir-377	HMDD v3.2
3	hsa-mir-34c	HMDD v3.2	13	hsa-mir-200a	HMDD v3.2
4	hsa-mir-562	HMDD v3.2	14	hsa-mir-1182	dbDEMC 3.0
5	hsa-mir-130b	HMDD v3.2	15	hsa-let-7g	HMDD v3.2
6	hsa-mir-495	HMDD v3.2	16	hsa-mir-526a-1	HMDD v3.2
7	hsa-mir-4298	HMDD v3.2	17	hsa-mir-106b	HMDD v3.2
8	hsa-mir-568	HMDD v3.2	18	hsa-mir-2355	HMDD v3.2
9	hsa-mir-23a	HMDD v3.2	19	hsa-mir-3130-2	HMDD v3.2
10	hsa-mir-378a	HMDD v3.2	20	hsa-mir-374a	HMDD v3.2

Colon tumor is one of the leading tumors in the world and it is considered among the big killers[27]. Early diagnosis of diseases is crucial for improving the survival rate of patients. Many studies have shown that a specific miRNA may exhibit the role of tumor suppressor gene or oncogene[28]. Table 3 shows that 19 of the top 20 miRNA could be corroborated by data from HMDDv3.2 and dbDEMC 3.0.

IV. CONCLUSION.

We propose a novel computational framework called TWMHGT based on the Multi-layer Heterogeneous Graph Transformer for predicting the sensitivity between miRNA and diseases. In this framework, we utilize the two-way MHGT model and concatenate its output embeddings to effectively integrate deep information. Finally, by weighting and fusing the two embedding outputs using attention mechanisms and decoding through matrix multiplication, we obtain the predicted miRNA-disease association information. Specifically, compared to previous methods, our model achieves higher values in terms of AUC, PRC, accuracy, sensitivity, and specificity. Additionally, case studies further validate the effectiveness of TWMHGT in predicting potential miRNA-disease associations. These findings indicate that TWMHGT can serve as a valuable tool for identifying miRNA-disease relationships, contributing to disease diagnosis and treatment. In future work, we will introduce other nodes related to miRNA-disease for meta-path construction to provide more effective information and strive for better results.

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